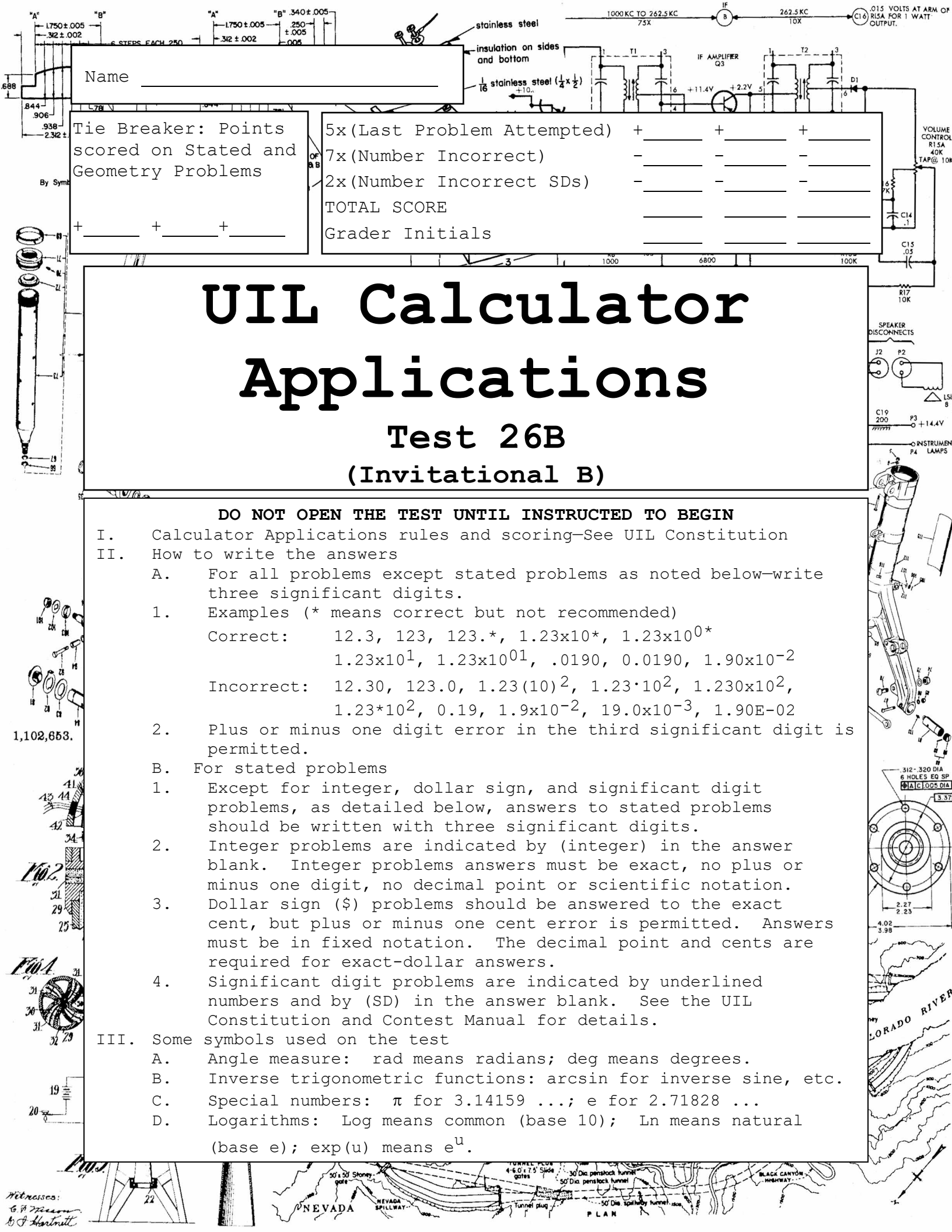




Test 26B
(Invitational B)

DO NOT OPEN THE TEST UNTIL INSTRUCTED TO BEGIN

- I. Calculator Applications rules and scoring—See UIL Constitution
- II. How to write the answers
- A. For all problems except stated problems as noted below—write three significant digits.
1. Examples (* means correct but not recommended)
- Correct: 12.3, 123, 123.*, $1.23 \times 10^*$, $1.23 \times 10^0^*$
 1.23×10^1 , 1.23×10^{01} , .0190, 0.0190, 1.90×10^{-2}
- Incorrect: 12.30, 123.0, $1.23(10)^2$, $1.23 \cdot 10^2$, 1.230×10^2 ,
 $1.23 \cdot 10^2$, 0.19, 1.9×10^{-2} , 19.0×10^{-3} , 1.90E-02
2. Plus or minus one digit error in the third significant digit is permitted.
- B. For stated problems
1. Except for integer, dollar sign, and significant digit problems, as detailed below, answers to stated problems should be written with three significant digits.
2. Integer problems are indicated by (integer) in the answer blank. Integer problems answers must be exact, no plus or minus one digit, no decimal point or scientific notation.
3. Dollar sign (\$) problems should be answered to the exact cent, but plus or minus one cent error is permitted. Answers must be in fixed notation. The decimal point and cents are required for exact-dollar answers.
4. Significant digit problems are indicated by underlined numbers and by (SD) in the answer blank. See the UIL Constitution and Contest Manual for details.
- III. Some symbols used on the test
- A. Angle measure: rad means radians; deg means degrees.
- B. Inverse trigonometric functions: arcsin for inverse sine, etc.
- C. Special numbers: π for 3.14159 ...; e for 2.71828 ...
- D. Logarithms: Log means common (base 10); Ln means natural (base e); $\exp(u)$ means e^u .

26B-1. $(\pi + 14.6) \times 6.44$ ----- 1= _____

26B-2. $(-0.0743 + 0.0266) \times (-0.295) - 0.0204$ ----- 2= _____

26B-3. $(-0.465 - 0.188 + 0.349) \times (-0.734) - 0.723$ ----- 3= _____

26B-4. $\frac{(6020 - 3020)}{\{(9.62)/(-8.5)\}} + (2150 - 1290)$ ----- 4= _____

26B-5. $\frac{\{(0.741 - 0.282 + 0.327)/(9.95)\}}{\{(1.32)(1.85)/(-4.9)\}}$ ----- 5= _____

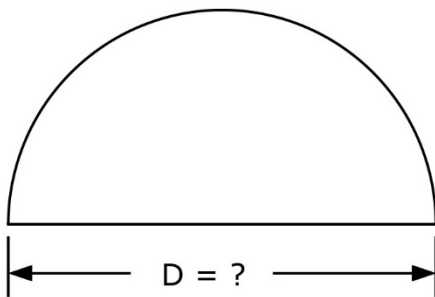
26B-6. Calculate the sum of 53.4 and 34.9. ----- 6= _____

26B-7. What is π raised to the power π , then added to 233? ----- 7= _____

26B-8. Calculate the remainder of 18.6 divided by 2.73. ----- 8= _____

26B-9.

SEMICIRCLE

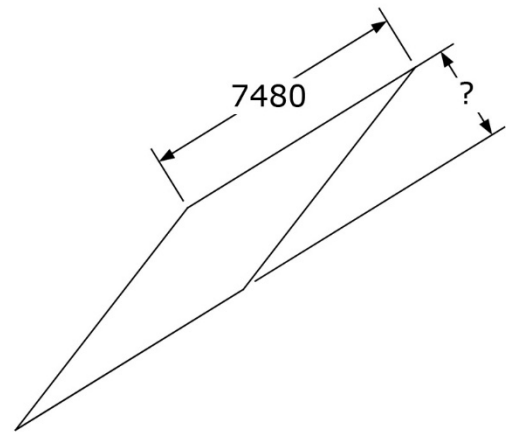


Perimeter = 82.3

26B-9 = _____

26B-10.

PARALLELOGRAM



Area = 2.05×10^7

26B-10 = _____

26B-11. $\frac{(4960 + 2220)}{(0.451 - 0.496)} + \frac{(-22600 + 92600)}{(4.26 - 3.22)}$ ----- 11=_____

26B-12. $\frac{(3.7)(29.1) - (-86.9 + 61.4)(-16.4)}{(-61.2 + 101 + 88.5)(26.9)}$ ----- 12=_____

26B-13. $\frac{(-0.441)(895 - 620)\{-0.034 - (-0.118)(0.111)\}}{(0.0909 + 0.0432)(0.437 - 0.462)}$ ----- 13=_____

26B-14. $\frac{\{(0.572 + 1.14)(1.58 + 7.21) + 16.6 - 12\}}{(-877 - 208)(-90.2 + 141 - 71)}$ ----- 14=_____

26B-15. $\frac{(93500 + 28800 - 34400)(0.206 - 0.178 - 0.214)}{(-9.58)(9.35)(3.71)(\pi + 2.05 + 2.66)}$ ----- 15=_____

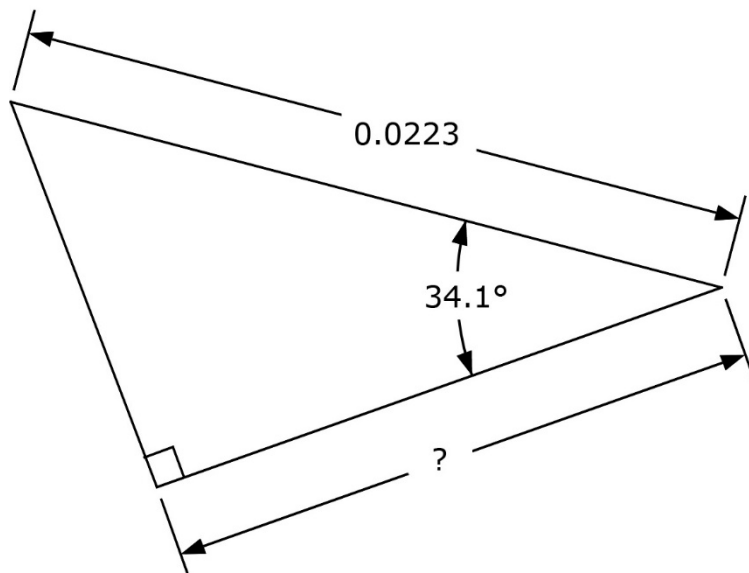
26B-16. Evan watches an average of 3 hr 16 min of TV daily. What fraction of waking time is spent watching TV? He averages 8 hr sleep. -----16=_____ %

26B-17. Mike has 35 quarters, 24 dimes, 59 nickels, and 134 pennies. How much money does he have? -----17=\$_____

26B-18. The High Roller Ferris Wheel in Las Vegas is 550 ft in diameter and makes one full revolution in a half hour. What is the velocity of the passenger enclosures? -----18=_____ mph

26B-19.

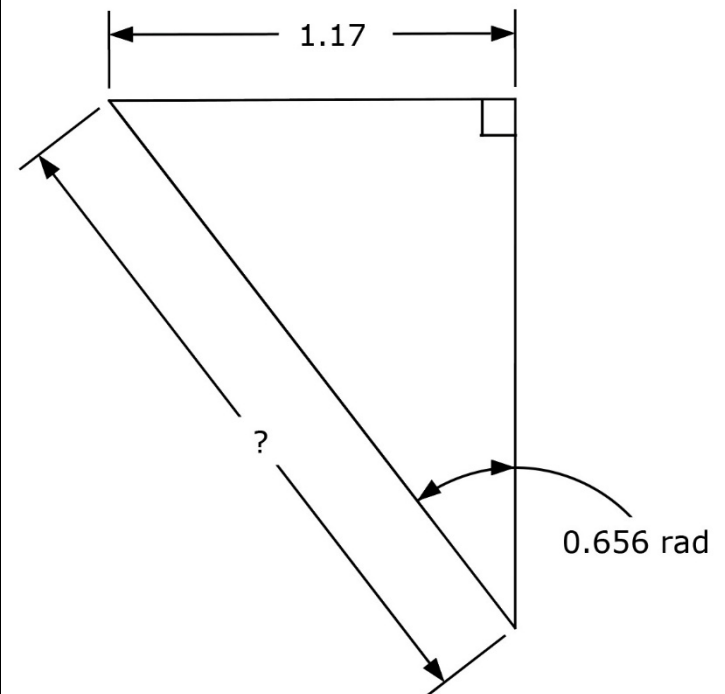
RIGHT TRIANGLE



26B-19 = _____

26B-20.

RIGHT TRIANGLE



26B-20 = _____

26B-21. $\left[\frac{\sqrt{2.25 - 0.419}}{-5.56} + \frac{(-1.58)}{7.46} \right]^2$ ----- 21=_____

26B-22. $\frac{1}{4.21 + 7.68} + \frac{1}{7.54 - 52.3} + \frac{1}{(2.88)}$ ----- 22=_____

26B-23. $\left[\frac{\pi + 0.604 + \sqrt{0.609/0.349}}{-13.5 + 22.2} \right]^2$ ----- 23=_____

26B-24. $\frac{\sqrt{12.1 + 9.24 + (1150)/(77)}}{8.96 + 7.21}$ ----- 24=_____

26B-25. $(0.399)(4.72) + \sqrt{(7.64)/(7.33)} + [(0.183)(5.61)]^2$ ----- 25=_____

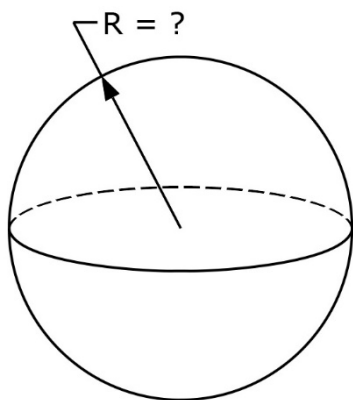
26B-26. The world's largest tree, "General Sherman", has a volume of 52,508 ft³. A wooden toothpick volume is 0.0050 in³. How many toothpicks could be made from the General Sherman, assuming 33% of the tree volume is lost to waste? -----26=_____ (SD)

26B-27. At what latitude, measured as an angle from the equator, is the earth's rotational surface velocity equal to 500 mph? -----27=_____ deg

26B-28. Wanda wants to net \$100 profit from a school donut sale. She buys a dozen donuts for \$11. If she sells the donuts for \$1.25 each, how many dozen donuts must she sell? -----28=_____ dozen (integer)

26B-29.

SPHERE

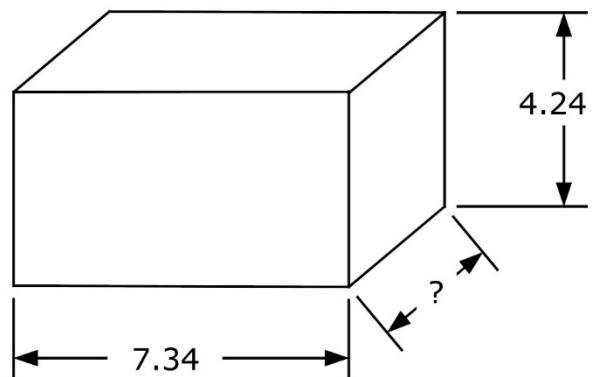


Total Surface Area = 0.0696

26B-29 = _____

26B-30.

CUBOID



Volume = 84.8

26B-30 = _____

$$26B-31. \left[\frac{-1.80 \times 10^6}{6.32 \times 10^6 + 3.44 \times 10^6} + 0.282 \right] \times \left\{ 803 + (-39.2)^2 - \sqrt{8.38 \times 10^6} \right\} \quad 31 = \underline{\hspace{2cm}}$$

$$26B-32. \frac{1}{3.28 \times 10^{-4}} + \frac{1}{\sqrt{1.40 \times 10^{-7}}} + \frac{(8.51 + 44.3 - 7.98)^2}{\sqrt{2.24 - 1.61}} \quad \text{-----} \quad 32 = \underline{\hspace{2cm}}$$

$$26B-33. \frac{\sqrt{(80800)/\{(52400)/\sqrt{14000}\}}}{5.34 + (0.506)(1.72)} + \{2.35 + 3.21\}^{1/2} \quad \text{-----} \quad 33 = \underline{\hspace{2cm}}$$

$$26B-34. \frac{(3.93)^2 + \sqrt{213}}{\sqrt{(0.00186)(-67.8)^2}} + \frac{\sqrt{\sqrt{(\pi)(0.167)}}}{0.0701 + 0.0828} \quad \text{-----} \quad 34 = \underline{\hspace{2cm}}$$

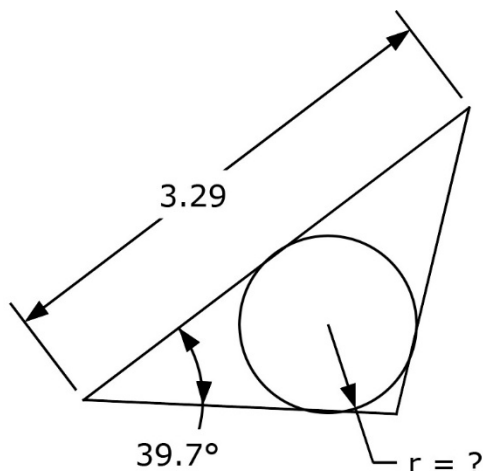
$$26B-35. \frac{\frac{1}{26200} + \frac{0.782}{(115 + 45.6)^2} - \frac{\sqrt{5.65 \times 10^{-12}}}{(-0.28)^2}}{(-0.954 + 1.26)^2 + (-0.247)} \quad \text{-----} \quad 35 = \underline{\hspace{2cm}}$$

26B-36. Larry and Gabe meet at a 440 yd oval track. They start running in opposite directions. When they meet again after 1 min, Larry had run 275 yd. What was Gabe's running speed? ----- 36 = mph

26B-37. A room has 210 people in it: boys, girls, men, and women. There 14 more females than males, 13 more boys than girls, and 28 more men than boys. How many girls were in the room? ----- 37 = integer

26B-38. Debby leaves Tyler driving 74 mi to Corsicana to the Collins St. Bakery. She travels at 55 mph. Megan in Waco wants to meet Debby in Corsicana for a snack, arriving at the bakery at the same time. She leaves Waco 19.7 min later than Debby and drives at 58 mph. What is the driving distance from Waco to Tyler? ----- 38 = mi

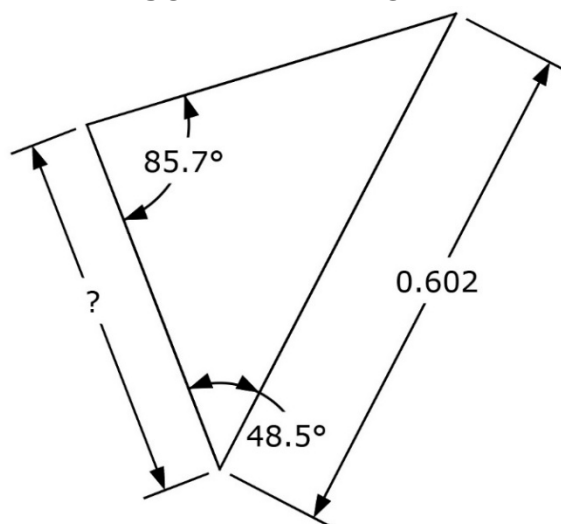
26B-39.
INSCRIBED ISOSCELES TRIANGLE



26B-39 =

26B-40.

SCALENE TRIANGLE



26B-40 =

26B-41. $\frac{10^{-(0.7 - 1.12)}}{-364 + 275}$ ----- 41=_____

26B-42. $-4.13 \times 10^5 e^{0.287} + (-83800) e^{-0.143}$ ----- 42=_____

26B-43. $\frac{56.6 - 77.6}{\text{Log}(52 + 28.1)}$ ----- 43=_____

26B-44. $(492 + 906)^{1/3} + 1/\{(308)^{-0.5}\}$ ----- 44=_____

26B-45.(deg) $\{(5.45 \times 10^7) \sin(-45^\circ)\} \times \{(-1.91 \times 10^7) \cos(-97^\circ)\}$ ----- 45=_____

26B-46. A farmer discovers that he can substitute four small chicken eggs for three regular-sized chicken eggs. What is the ratio of the egg diameters, a number less than one? -----46=_____

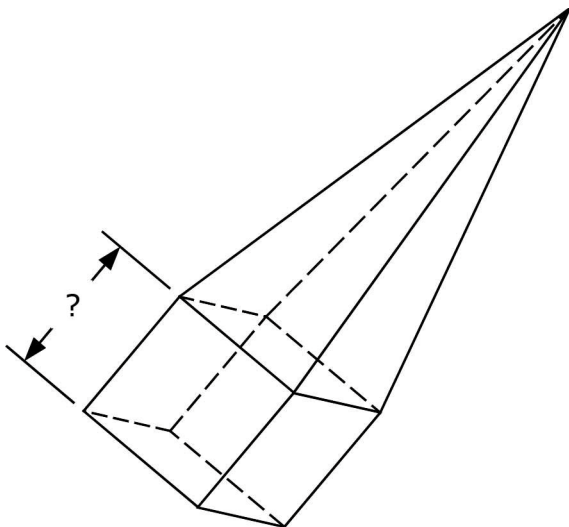
26B-47. What is the correlation coefficient for these data: (1, 14), (2, 29), (3, 31), (4, 42), (5, 50)? -----47=_____

26B-48. For what value of x between 0 and 1 does $3^x = 5x$?-----48=_____

26B-49.

CUBE AND SQUARE PYRAMID

Volume(Cube) = Volume(Pyramid)

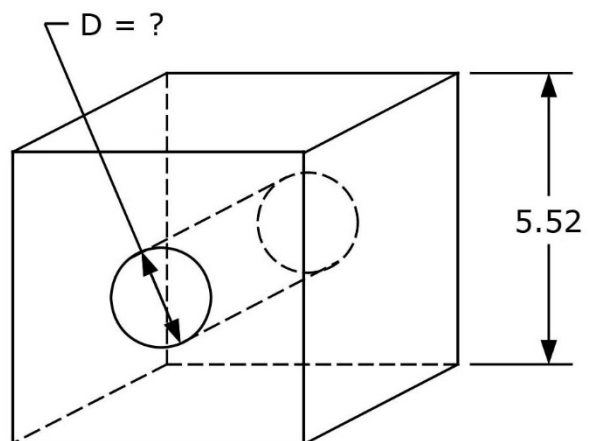


Total Surface Area = 993

26B-49 = _____

26B-50.

CUBE WITH CYLINDRICAL HOLE



Volume = 152

26B-50 = _____

26B-51. $\frac{10^{(0.563)} \times 10^{-(0.591)} + 0.79}{10^{(1.15 + 0.336)}} \text{ ----- } 51 = \underline{\hspace{2cm}}$

26B-52. $\frac{41.2 + e^{(1.93 + 1.8)}}{0.955 - e^{-(0.21 - 0.654)}} \text{ ----- } 52 = \underline{\hspace{2cm}}$

26B-53. $(-0.0038) \ln \left[\frac{0.00479 + (0.00585)(0.508)}{0.00708 + 0.0125} \right] \text{ ----- } 53 = \underline{\hspace{2cm}}$

26B-54. $\frac{(\pi)^{0.204} - (7.8)^{-0.24}}{80200 + 13400} \text{ ----- } 54 = \underline{\hspace{2cm}}$

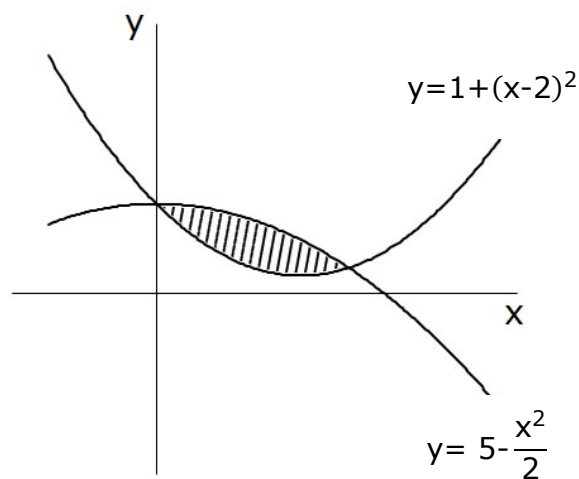
26B-55.(rad) $\arctan \left[\frac{(2930)(0.398)}{(1.46)(43.8)} \right] + (0.443)(1.47) \text{ ----- } 55 = \underline{\hspace{2cm}}$

26B-56. (rad) Calculate the area under the curve $y = 3\cos[\pi(x-2.5)/5]$ between 0 and 5. ----- 56 =

26B-57. A line of negative slope passes through the point (6,3.5). It thus forms a right triangle with sides equal to the line intersections with the x-axis and y-axis, and the negative-sloped line segment forms the hypotenuse. What is the ratio of side dimensions, a number greater than one, that maximizes the right triangle area? ----- 57 =

26B-58. Calculate c if the determinant of $\begin{bmatrix} 1.1 & 6c & 7 \\ 4.6 & 5.7 & 5 \\ 7 & 1 & 9.8 \end{bmatrix} = -20$. ----- 58 =

26B-59.



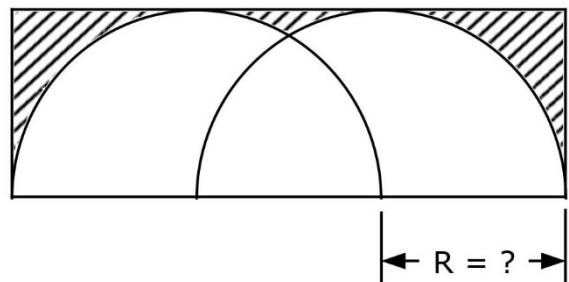
Shaded Area = ?

26B-59 =

26B-60.

RECTANGLE AND IDENTICAL SEMICIRCLES

Shaded Area = 2.35



26B-60 =

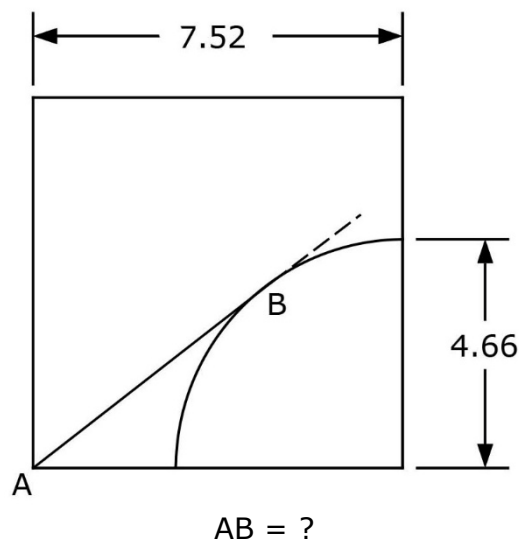
26B-61. Light intensity is directly proportional to the bulb wattage and inversely proportional to the square of the distance from the bulb. Brenda has adequate light to read at a distance of 6 ft from a 150 watt bulb. How far does she need to move toward the bulb if she changed it out with a 100 watt bulb? -----61=_____ft

26B-62. Calculate 7.56×10^{45} raised to the power -282 . -----62=_____

26B-63. An archery target is 230 ft away. Larry shoots an arrow directly at the target with a velocity of 200 ft/s. What is the percent error in the time of flight, if he assumes the arrow travels in a straight line rather than on a trajectory? -----63=_____%

26B-64.

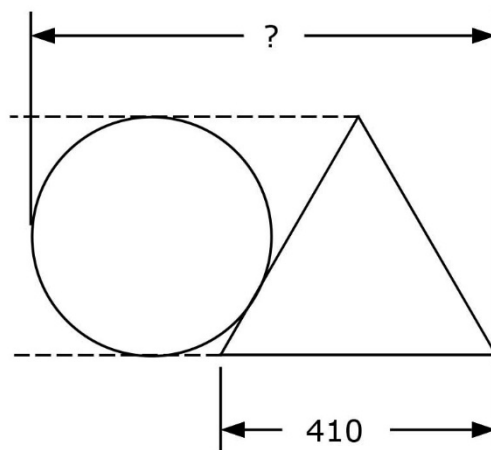
SQUARE AND QUARTER CIRCLE



26B-64 = _____

26B-65.

CIRCLE AND EQUILATERAL TRIANGLE



26B-65 = _____

26B-66. $\frac{\sqrt{e^{-(0.778 + 0.575)}}}{\{e^{(0.778 - 0.934)}\}^2} \times \sqrt[3]{(4.79)^2}$ ----- 66=_____

26B-67. (rad) $\cos(1.19 - 0.134) - \cos(1.19 + 0.134)$ ----- 67=_____

26B-68. (rad) $\frac{1}{(4950)(0.111)} \ln\{(7.78) + (-5.28)\sin(1.58)\}$ ----- 68=_____

26B-69. $1 + 0.64 + (0.64)^2 + \frac{(0.64)^4}{8} - \frac{(0.64)^5}{15}$ ----- 69=_____

26B-70. $\frac{(-0.463)}{(3.85)} - \frac{(3.51)}{(8.11)^2} \ln\left[\frac{(-0.0611)^2 + (0.00272)}{(0.463) + \sqrt{0.553}}\right]$ ----- 70=_____

26B-1	= 114 = 1.14×10^2	26B-11	= -92200 = -9.22×10^4	26B-21	= 0.207 = 2.07×10^{-1}
26B-2	= -0.00633 = -6.33×10^{-3}	26B-12	= -0.0900 = -9.00×10^{-2}	26B-22	= 0.409 = 4.09×10^{-1}
26B-3	= -0.500 = -5.00×10^{-1}	26B-13	= -756 = -7.56×10^2	26B-23	= 0.339 = 3.39×10^{-1}
26B-4	= -1790 = -1.79×10^3	26B-14	= 0.000896 = 8.96×10^{-4}	26B-24	= 0.372 = 3.72×10^{-1}
26B-5	= -0.159 = -1.59×10^{-1}	26B-15	= 6.27 = 6.27×10^0	26B-25	= 3.96 = 3.96×10^0
26B-6	= 88.3 = 8.83×10^1	26B-16	= 20.4 = 2.04×10^1	26B-26	= 1.2×10^{10} (2SD)
26B-7	= 269 = 2.69×10^2	26B-17	= \$15.44	26B-27	= 61.2 = 6.12×10^1
26B-8	= 2.22 = 2.22×10^0	26B-18	= 0.654 = 6.54×10^{-1}	26B-28	= 25 integer
26B-9	= 32.0 = 3.20×10^1	26B-19	= 0.0185 = 1.85×10^{-2}	26B-29	= 0.0744 = 7.44×10^{-2}
26B-10	= 2740 = 2.74×10^3	26B-20	= 1.92 = 1.92×10^0	26B-30	= 2.72 = 2.72×10^0

26B-31	= -54.2 = -5.42x10 ¹	26B-41	= -0.0296 = -2.96x10 ⁻²	26B-51	= 0.0564 = 5.64x10 ⁻²	26B-61	= 1.10 = 1.10x10 ⁰
26B-32	= 8250 = 8.25x10 ³	26B-42	= -623000 = -6.23x10 ⁵	26B-52	= -137 = -1.37x10 ²	26B-62	= 1.81x10 ⁻¹² ,938
26B-33	= 4.53 = 4.53x10 ⁰	26B-43	= -11.0 = -1.10x10 ¹	26B-53	= 0.00352 = 3.52x10 ⁻³	26B-63	= -0.432 = -4.32x10 ⁻¹
26B-34	= 15.8 = 1.58x10 ¹	26B-44	= 28.7 = 2.87x10 ¹	26B-54	= 6.97x10 ⁻⁶	26B-64	= 5.90 = 5.90x10 ⁰
26B-35	= -0.000249 = -2.49x10 ⁻⁴	26B-45	= -8.97x10 ¹³	26B-55	= 2.17 = 2.17x10 ⁰	26B-65	= 690 = 6.90x10 ²
26B-36	= 5.63 = 5.63x10 ⁰	26B-46	= 0.909 = 9.09x10 ⁻¹	26B-56	= 9.55 = 9.55x10 ⁰	26B-66	= 1.97 = 1.97x10 ⁰
26B-37	= 22 integer	26B-47	= 0.981 = 9.81x10 ⁻¹	26B-57	= 1.71 = 1.71x10 ⁰	26B-67	= 0.248 = 2.48x10 ⁻¹
26B-38	= 133 = 1.33x10 ²	26B-48	= 0.269 = 2.69x10 ⁻¹	26B-58	= -2.83 = -2.83x10 ⁰	26B-68	= 0.00167 = 1.67x10 ⁻³
26B-39	= 0.594 = 5.94x10 ⁻¹	26B-49	= 9.47 = 9.47x10 ⁰	26B-59	= 4.74 = 4.74x10 ⁰	26B-69	= 2.06 = 2.06x10 ⁰
26B-40	= 0.433 = 4.33x10 ⁻¹	26B-50	= 1.93 = 1.93x10 ⁰	26B-60	= 2.23 = 2.23x10 ⁰	26B-70	= 0.159 = 1.59x10 ⁻¹